The components of the Phillips curve in a New-Keynesian gap model

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Closed economy

The Phillips curve

- The Phillips curve of a closed economy relates the real marginal cost gap to inflation

$$\pi_t = \beta_1 \pi_{t+1} + (1 - \beta_1) \pi_{t-1} + \beta_2 \tilde{rmc}_t \tag{1}$$

- But what determines the RMC gap?
- In particular, can we relate the RMC gap to other variables in the model, such as output gap, RER gap, etc.?

Production with labor

- A simple model
  - Take the simplified NK model, with labor as the only production factor
  - But keep everything else, including the constant returns to scale production function
  - Then we showed that the RMC gap is proportional to the output gap
- The relevant equations

$$\phi\, \tilde{l}_t = -\sigma\, \tilde{c}_t + \tilde{wr}_t \tag{2}$$

$$\tilde{wr}_t = \tilde{y}_t - \tilde{l}_t + \tilde{rmc}_t \tag{3}$$

$$\tilde{y}_t = \tilde{c}_t \tag{4}$$

$$\tilde{y}_t = \tilde{a}_t + \tilde{l}_t \tag{5}$$

RMC and output gap

- The calculation

$$\begin{aligned} \tilde{rmc}_t &= \tilde{wr}_t - (\tilde{y}_t - \tilde{l}_t) = \phi\,\tilde{l}_t + \sigma\,\tilde{c}_t - \tilde{a}_t = \\ &= \phi\,(\tilde{y}_t - \tilde{a}_t) + \sigma\,\tilde{y}_t - \tilde{a}_t = \\ &= (\sigma + \phi)\left(\tilde{y}_t - \frac{1+\phi}{\sigma+\phi}\,\tilde{a}_t\right) \end{aligned} \tag{6}$$

- When prices are flexible, the RMC gap is zero

$$0 = (\sigma + \phi)\left(\tilde{y}_t^f - \frac{1+\phi}{\sigma+\phi}\,\tilde{a}_t\right) \tag{7}$$

- And so the difference is

$$\tilde{rmc}_t = (\sigma + \phi)\left(\tilde{y}_t - \tilde{y}_t^f\right) \tag{8}$$

Assumptions

- The derivation relies on
  - The production function $y_t = A_t\, l_t$
  - The form of RMC, which we know is derived from constant returns to scale

$$rmc_t = \frac{wr_t}{(1-\alpha)\,\frac{y_t}{l_t}} \tag{9}$$

  - The labor supply function of households, which implies that labor is anticyclical

Open economy

External factors

- What happens if we open up our economy?
- Import prices will affect CPI
  - Either directly, through imported consumption goods
  - Or indirectly, through intermediate imports
- The first channel will give us an immediate one-to-one response in import prices to foreign (and exchange rate) shocks
- The second one will result in longer pass-through
- We can handle important external goods (i.e. food) separately

Setup and notation

- Let us fix some notation before we continue
  - There is a domestically produced good whose price we denote by $p_{t,D}$
  - There is a foreign good, whose price *in foreign currency* we denote by p^*
  - A fraction of the imported foreign good might be *directly consumed*, in which case its price *in domestic currency* we denote by $p_{t,Fd}$
  - The remaining fraction is further processed, and final consumer price *in domestic currency* is denoted by $p_{t,F}$
  - Consumer inflation (in log-difference) is the weighed average of the three component

$$\pi_t = (1 - \gamma_{Fd} - \gamma_F)\, \pi_{t,D} + \gamma_{Fd}\, \pi_{t,Fd} + \gamma_F\, \pi_{t,F} \tag{10}$$

  - The nominal exchange rate will be denoted by e_t, which is the domestic price of 1 unit of foreign currency, so depreciation is up
  - The real exchange rate is given as $z_t = \frac{e_t\, p_t^*}{p_t}$

Open economy

Indirect effects

Direct effects

Overall Phillips curve

Food prices

Importers

- There are foreign price effects that take place slowly
- To model this phenomenon, we assume that there are firms which import foreign consumption goods, and transform them into domestic consumption goods
- These firms behave exactly as domestic basic good producers
  - They are monopolistically competitive, and charge a markup
  - They face menu costs, so their price will be different from the "optimal" one, given by marginal cost
  - The low of one price "holds at the docks", meaning that the price they pay for the imported goods is $e_t\, p_t^*$

Foreign price Phillips curve

- The first two assumptions lead to a separate Phillips curve for *indirect foreign inflation*

$$\pi_{t,F} = \beta_1 \, \pi_{t+1,F} + (1 - \beta_1) \, \pi_{t-1,F} + \beta_2 \, r\tilde{m}c_{t,F} \tag{11}$$

- And the last assumption implies that the real marginal cost of the importing firms is the real exchange rate

$$r\tilde{m}c_{t,F} = \tilde{z}_t \tag{12}$$

Open economy

Indirect effects

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Direct imported inflation

- For some imported goods, we may assume that the *law of one price holds* for their *final consumer price*
- We will call these goods *direct imported goods*, so that

$$p_{t,Fd} = e_t\, p_t^* \tag{13}$$

- Using the definition of the real exchange $z_t = \frac{e_t\, p_t^*}{p_t}$, we get that

$$p_{t,Fd} = \frac{e_t\, p_t^*}{p_t}\, p_t = z_t\, p_t \tag{14}$$

- In log-linear approximation

$$\pi_{t,Fd} = \Delta z_t + \pi_t \tag{15}$$

Open economy

- Indirect effects
- Direct effects
- Overall Phillips curve
- Food prices

Phillips curve with imported inflation

- The Phillips curve will then be

$$\begin{aligned}\pi_t &= (1-\gamma_{Fd}-\gamma_F)\,\pi_{t,D} + \gamma_{Fd}\,\pi_{t,Fd} + \gamma_F\,\pi_{t,F} \\ &= (1-\gamma_{Fd}-\gamma_F)\left(\beta_1\,\pi_{t+1,D} + (1-\beta_1)\,\pi_{t-1,D} + \beta_2\,r\tilde{m}c_{t,D}\right) \\ &\quad + \gamma_{Fd}\left(\Delta z_t + \pi_t\right) \\ &\quad + \gamma_F\left(\beta_1\,\pi_{t+1,F} + (1-\beta_1)\,\pi_{t-1,F} + \beta_2\,r\tilde{m}c_{t,F}\right)\end{aligned} \tag{16}$$

Trend adjustments

- But there's a problem here: the real exchange rate trend depreciation appears directly in the Phillips curve
- We rather think that trends are connected to trends (i.e. target) inflation
- So we simply remove the RER trend depreciation
- Note that as of this step we are not really "deriving" equations, we are rather "motivating" a useful form of the open economy Phillips curve
  - After all, the gap model is not a DSGE model

De-trended direct foreign inflation

- We want to have a term that represents direct imported inflation
- This should be free of real trends growth rates
- So we (re)define direct imported inflation as

$$\pi_{t,Fd} = \pi_t + \Delta z_t - \Delta \bar{z}_t \tag{17}$$

- This variable only has the nominal trend (e.g. the inflation target) growth
- The formula can be re-arranged to have something more familiar

$$\begin{aligned} \pi_{t,Fd} &= \pi_t + \Delta z_t - \Delta \bar{z}_t \\ &= \Delta e_t + \pi^*_t - \pi_t + \pi_t - \Delta \bar{z}_t \\ &= \pi^*_t + \Delta e_t - \Delta \bar{z}_t \end{aligned} \tag{18}$$

The de-trended Phillips curve

- We get after substitution

$$\begin{aligned}\pi_t &= (1-\gamma_{Fd}-\gamma_F)\,\pi_{t,D} + \gamma_{Fd}\,\pi_{t,Fd} + \gamma_F\,\pi_{t,F} \\ &= (1-\gamma_{Fd}-\gamma_F)\left(\beta_1\,\pi_{t+1,D} + (1-\beta_1)\,\pi_{t-1,D} + \beta_2\,r\tilde{m}c_{t,D}\right) \\ &\quad + \gamma_{Fd}\left(\pi_t^* + \Delta e_t - \Delta\bar{z}_t\right) \\ &\quad + \gamma_F\left(\beta_1\,\pi_{t+1,F} + (1-\beta_1)\,\pi_{t-1,F} + \beta_2\,r\tilde{m}c_{t,F}\right)\end{aligned} \tag{19}$$

- Here we make another simplifying assumption: we replace lags/leads of the individual components on the right-hand-side with lags/leads of overall inflation π_t

The simplified Phillips curve

- We get after this simplification that

$$\begin{aligned}\pi_t = {} & (1-\gamma_{Fd}-\gamma_F)\left(\beta_1\,\pi_{t+1}+(1-\beta_1)\,\pi_{t-1}+\beta_2\,r\tilde{m}c_{t,D}\right)\\ & +\gamma_{Fd}\left(\pi_t^*+\Delta e_t-\Delta\bar{z}_t\right)\\ & +\gamma_F\left(\beta_1\,\pi_{t+1}+(1-\beta_1)\,\pi_{t-1}+\beta_2\,r\tilde{m}c_{t,F}\right)\end{aligned} \tag{20}$$

- And now we combine lags/leads to get

$$\begin{aligned}\pi_t = {} & (1-\gamma_{Fd})\,\beta_1\,\pi_{t+1}+(1-\gamma_{Fd})\,(1-\beta_1)\,\pi_{t-1}\\ & +\gamma_{Fd}\left(\pi_t^*+\Delta e_t-\Delta\bar{z}_t\right)\\ & +\left((1-\gamma_{Fd}-\gamma_F)\,\beta_2\,r\tilde{m}c_{t,D}+\gamma_F\,\beta_2\,r\tilde{m}c_{t,F}\right)\end{aligned} \tag{21}$$

Overall RMC

- We rewrite the marginal cost effects as

$$\begin{aligned}
&(1-\gamma_{Fd}-\gamma_F)\,\beta_2\,\tilde{rmc}_{t,D} + \gamma_F\,\beta_2\,\tilde{rmc}_{t,F} \\
&= (1-\gamma_{Fd})\,\beta_2\left(\frac{(1-\gamma_{Fd}-\gamma_F)\,\beta_2}{(1-\gamma_{Fd})\,\beta_2}\,\tilde{rmc}_{t,D} + \frac{\gamma_F\,\beta_2}{(1-\gamma_{Fd})\,\beta_2}\,\tilde{rmc}_{t,F}\right) \\
&= (1-\gamma_{Fd})\,\beta_2\left(\frac{1-\gamma_{Fd}-\gamma_F}{1-\gamma_{Fd}}\,\tilde{rmc}_{t,D} + \frac{\gamma_F}{1-\gamma_{Fd}}\,\tilde{rmc}_{t,F}\right) \\
&= (1-\gamma_{Fd})\,\beta_2\left(\left(1-\frac{\gamma_F}{1-\gamma_{Fd}}\right)\tilde{rmc}_{t,D} + \frac{\gamma_F}{1-\gamma_{Fd}}\,\tilde{rmc}_{t,F}\right)
\end{aligned} \tag{22}$$

- Which leads to the definition of the overall real marginal cost as

$$\tilde{rmc}_t = \left(1-\frac{\gamma_F}{1-\gamma_{Fd}}\right)\tilde{rmc}_{t,D} + \frac{\gamma_F}{1-\gamma_{Fd}}\,\tilde{rmc}_{t,F} \tag{23}$$

The final Phillips curve

- The final form of the Phillips curve will be

$$\begin{aligned}\pi_t = {} & (1-\gamma_{Fd})\,\beta_1\,\pi_{t+1} + (1-\gamma_{Fd})\,(1-\beta_1)\,\pi_{t-1} \\ & + \gamma_{Fd}\left(\pi_t^* + \Delta e_t - \Delta\overline{z}_t\right) + (1-\gamma_{Fd})\,\beta_2\,r\tilde{m}c_t\end{aligned} \tag{24}$$

where

$$r\tilde{m}c_t = \left(1 - \frac{\gamma_F}{1-\gamma_{Fd}}\right) r\tilde{m}c_{t,D} + \frac{\gamma_F}{1-\gamma_{Fd}}\, r\tilde{m}c_{t,F} \tag{25}$$

- So the Phillips curve will be homogeneous in three components: expected inflation, lagged inflation, and direct imported inflation
- All three will have the same steady state equal to the inflation target
- The real marginal cost gap is a weighted average of domestic and foreign costs

Open economy

Indirect effects

Direct effects

Overall Phillips curve

Food prices

General considerations

- We want to a separate Phillips curve for food prices
- The preceding arguments can be repeated verbatim
- There's only one caveat: whenever foreign prices are used, it must be foreign food prices
- Having several price indices may lead to complications with relative prices
  - Foreign food prices might growth at a different rate than foreign CPI (even in the long run)
  - Domestic food prices might growth at a different rate than domestic CPI (even in the long run)
- The long run trends in the relative prices must be correctly captured, to ensure accurate forecasting performance
- But without jeopardizing the CB-s *aggregate CPI* target

Trend and gap adjustments

- One possible solution to these problems is to adjust terms using trend relative prices
- This is similar in spirit to the adjustment of direct imported inflation by the real exchange rate trend
- Other adjustments are necessary because we do not want to change the definition of the real exchange rate
  - These adjustments ensure that imported price pressures are correctly captured by comparing foreign and domestic food prices, even if the RER is defined using core prices
- For the same reason, gap adjustments will also be necessary
- The solution presented here, there are other options as well

Relative prices

- We define foreign and domestic relative food price as follows

$$rp_t^{*,\,\text{food}} = p_t^{*,\,\text{food}} - p_t^* \tag{26}$$
$$rp_t^{\text{food}} = p_t^{\text{food}} - p_t \tag{27}$$

  where p_t is *aggregate CPI* level

- The second definition implicitly introduces the relative core price

$$rp_t^{\text{core}} = p_t^{\text{core}} - p_t \tag{28}$$

- As usual for real variables, we decompose the relative prices as the sum of a gap and a trend

$$rp_t^{*,\,\text{food}} = \tilde{rp}_t^{*,\,\text{food}} + \overline{rp}_t^{*,\,\text{food}} \tag{29}$$
$$rp_t^{\text{food}} = \tilde{rp}_t^{\text{food}} + \overline{rp}_t^{\text{food}} \tag{30}$$
$$rp_t^{\text{core}} = \tilde{rp}_t^{\text{core}} + \overline{rp}_t^{\text{core}} \tag{31}$$

The food Phillips curve I

- Let's start from the core Phillips curve

$$\pi_t^{\text{core}} = (1-\gamma_{Fd})\,\beta_1\,\pi_{t+1}^{\text{core}} + (1-\gamma_{Fd})\,(1-\beta_1)\,\pi_{t-1}^{\text{core}} + \gamma_{Fd}\left(\pi_t^{*} + \Delta e_t - \Delta\bar{z}_t\right) + (1-\gamma_{Fd})\,\beta_2\,r\tilde{m}c_t \qquad (32)$$

where

$$r\tilde{m}c_t = \left(1 - \frac{\gamma_F}{1-\gamma_{Fd}}\right) r\tilde{m}c_{t,D} + \frac{\gamma_F}{1-\gamma_{Fd}}\, r\tilde{m}c_{t,F} \qquad (33)$$

- We'll use the same parameter names, but of course values will be different from those for core inflation

The food Phillips curve II

- Specifically for Rwanda, we will assume that the domestic component of RMC is irrelevant, so

$$\tilde{rmc}_t = \tilde{rmc}_{t,F} = \tilde{z}_t \tag{34}$$

- Furthermore, the real exchange rate will still be defined using core prices

$$z_t = e_t + p_t^* - p_t^{\text{core}} \tag{35}$$

- This is the definition which most accurately captures the effects of real convergence

The food Phillips curve III

- We first have to replace π_t^* with $\pi_t^{*,\,\text{food}}$ in the direct imported inflation term

$$\pi_t^* + \Delta e_t - \Delta\bar{z}_t \Longrightarrow \pi_t^{*,\,\text{food}} + \Delta e_t - \Delta\bar{z}_t \tag{36}$$

- We have to adjust for the foreign relative price trend, since the RER is defined using foreign CPI

$$\pi_t^{*,\,\text{food}} + \Delta e_t - \Delta\bar{z}_t \Longrightarrow \pi_t^{*,\,\text{food}} - \Delta\overline{rp}_t^{*,\,\text{food}} + \Delta e_t - \Delta\bar{z}_t \tag{37}$$

- The adjustment removes the foreign relative food price trend

The food Phillips curve IV

- The next adjustment is necessary since the RER is defined with domestic core inflation

$$\pi_t^{*,\text{ food}} - \Delta\overline{rp}_t^{*,\text{ food}} + \Delta e_t - \Delta\bar{z}_t \\ \Longrightarrow \\ \pi_t^{*,\text{ food}} - \Delta\overline{rp}_t^{*,\text{ food}} + \Delta e_t - \Delta\bar{z}_t - \Delta\overline{rp}_t^{\text{core}} \tag{38}$$

- The adjustment removes the domestic relative core price trend (but adds domestic relative food price trend)

The food Phillips curve V

- We now have a term which compares foreign food prices to domestic CPI
- The final trend adjustment ensures that we are comparing to domestic food prices

$$\pi_t^{*,\text{ food}} - \Delta\overline{rp}_t^{*,\text{ food}} + \Delta e_t - \Delta\bar{z}_t - \Delta\overline{rp}_t^{\text{core}} \\ \Longrightarrow \\ \pi_t^{*,\text{ food}} - \Delta\overline{rp}_t^{*,\text{ food}} + \Delta e_t - \Delta\bar{z}_t - \Delta\overline{rp}_t^{\text{core}} + \Delta\overline{rp}_t^{\text{food}} \tag{39}$$

- The adjustment removes the domestic relative food price trend

The food Phillips curve VI

- What's left is to apply the same logic to the RMC gap
- This gives

$$\tilde{z}_t \Longrightarrow \tilde{rp}_t^{*,\,\text{food}} + \tilde{z}_t + \tilde{rp}_t^{\text{core}} - \tilde{rp}_t^{\text{food}} \tag{40}$$

The food Phillips curve VII

- The final form of the food Phillips curve will therefore be

$$
\begin{aligned}
\pi_t^{\text{food}} = {} & (1-\gamma_{Fd})\,\beta_1\,\pi_{t+1}^{\text{food}} + (1-\gamma_{Fd})\,(1-\beta_1)\,\pi_{t-1}^{\text{food}} \\
& + \gamma_{Fd}\left(\pi_t^{*,\text{ food}} - \Delta\overline{rp}_t^{*,\text{ food}} + \Delta e_t - \Delta\bar{z}_t - \Delta\overline{rp}_t^{\text{core}} + \Delta\overline{rp}_t^{\text{food}}\right) \\
& + (1-\gamma_{Fd})\,\beta_2\left(\tilde{rp}_t^{*,\text{ food}} + \tilde{z}_t + \tilde{rp}_t^{\text{core}} - \tilde{rp}_t^{\text{food}}\right)
\end{aligned}
$$

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